natera™ | Signatera™ Residual disease test (MRD)

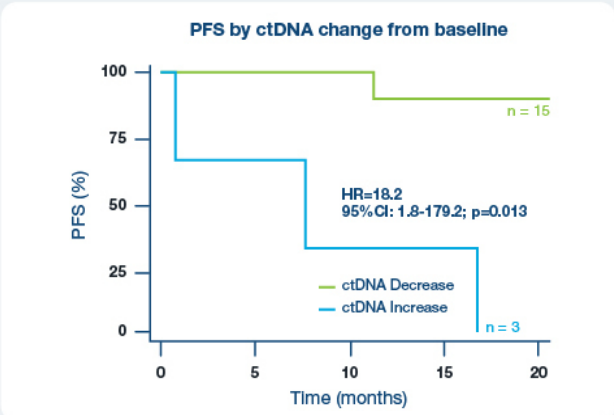
Covered by Medicare

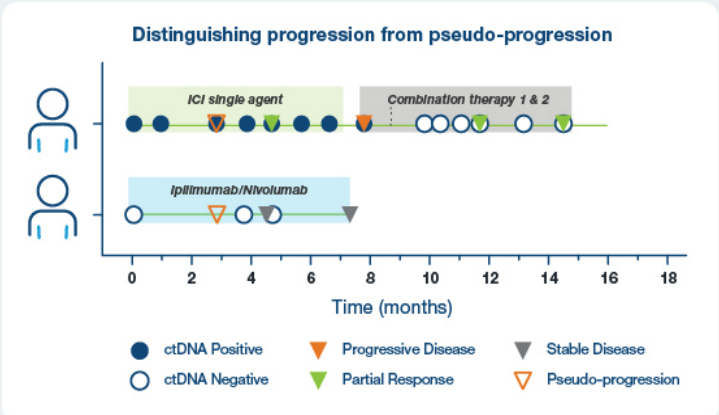
METASTATIC MELANOMA

Use Signatera™ ctDNA dynamics to inform earlier treatment decisions in metastatic melanoma patients

Early on-treatment ctDNA dynamics were predictive of PFS in metastatic melanoma patients receiving 1st line ICI treatment[1]

At week 6, Signatera™ identified that patients with increasing ctDNA had a 18x higher risk of progression than ctDNA-negative patients

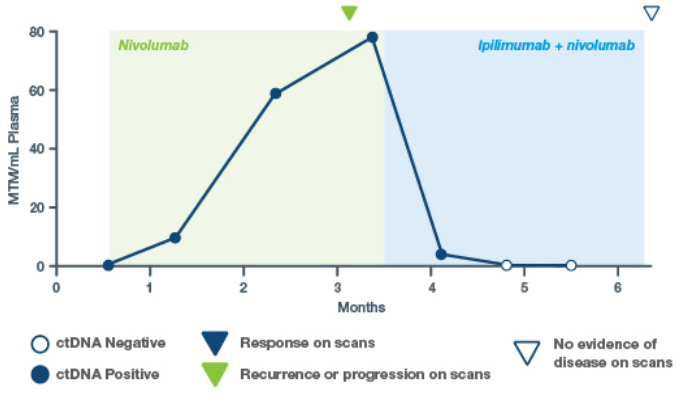




- Patients with any increase in ctDNA levels from baseline by week 6 of 1st Line ICI treatment (monotherapy and combination ICIs) had a significantly shorter PFS (HR: 18; p=0.013).
- Signatera™ was able to help distinguish between true vs pseudo-progression

Should treatment be changed or escalated?

Early rise in ctDNA can help inform treatment escalation or change



60-year-old stage IV metastatic melanoma patient[2]

Adapted from a real-world Signatera™ patient case. Case features modified to protect patient confidentiality. No treatment recommendations are made or should be implied.

Learn more at **www.natera.com/signatera-skincancers-clinicians**

Covered by Medicare for immunotherapy treatment response monitoring across all stages for solid tumors

PFS = Progression-free survival

References:

1. Eroglu Z, Krinshpun S, Kalashnikova E, et al. Circulating tumor DNA based molecular residual disease detection for treatment monitoring in advanced melanoma patients. *Cancer* (2023). https://doi.org/10.1002/cncr.34716
2. Eroglu Z, Krinshpun S, Martin J, et al. Molecular residual disease detection and circulating tumor DNA dynamics during treatment in patients with advanced melanoma. Abstract presented at: 10th World Congress of Melanoma; April 15-17, 2021; virtual meeting. #P-096. https://worldmelanoma2021.com/

13011 McCallen Pass, Building A Suite 100 | Austin, TX 78753 | natera.com

Signatera has been developed and its performance characteristics determined by the CLIA-certified laboratory performing the test. The test has not been cleared or approved by the US Food and Drug Administration (FDA). CAP accredited, ISO 13485 certified, and CLIA certified.

© 2023 Natera, Inc. All Rights Reserved. SGN_AD_MelanomaIO_20230725_NAT-8021346

natera™

Review Article

Inclusion of Minorities and Women in Cancer Clinical Trials, a Decade Later: Have We Improved?

Kat Kwiatkowski, MPH[1]; Kathryn Coe, PhD[1]; John C. Bailar, MD[2]; and G. Marie Swanson, PhD, MPH[1]

BACKGROUND: Inclusion of diverse groups of participants in cancer clinical trials is an important methodological and clinical issue. The quality of the science and generalizability of results depends on the inclusion of study participants who represent all populations among whom these treatment and prevention approaches will be used. **METHODS:** We conducted a systematic review using OVID as the primary source of reports included. Based on 304 peer-reviewed publications, diversity in the inclusion and reporting of study participants during a decade of cancer treatment and prevention trials (2001-2010) is summarized. Recommendations are made for improvements in the science and reporting of cancer clinical trials. **RESULTS:** Of the 277 treatment trials and 27 prevention trials included in this report, more than 80% of participants were white and 59.8% were male. In the recent decade, race and sex are rarely used as selection criteria unless the trial is focused on a sex-specific cancer. **CONCLUSIONS:** Women and racial/ethnic minorities remain severely underrepresented in cancer clinical trials, thus limiting the generalizability of cancer clinical research. ***Cancer* 2013;119:2956-63.** *© 2013 American Cancer Society.*

KEYWORDS: cancer clinical trials; treatment; prevention; health disparities; participant selection.

INTRODUCTION

In 2002, Swanson and Bailar published a widely cited paper on the enrollment of women and minorities in cancer prevention and treatment clinical trials during 1990-2000.[1] Their results were not encouraging. We report an update of that study, focused on identifying changes in enrollment in cancer treatment and prevention clinical trials a decade later (2001-2010).

It is understood that cancer treatment and preventive efforts are not generalizable to the entire treatable population. Historically, however, the majority of studies assessing the effectiveness of cancer treatment and prevention has been conducted using white men.[1] An improved understanding of the differences between subgroups (race/ethnicity and sex) is critical to improving the risk/benefit profile for a wide range of chemotherapeutics and prevention efforts.

Though much effort had been devoted to planning and implementation of programs aimed at reducing health disparities, the 2002 paper reported that minorities were still unlikely to be enrolled in clinical trials. There were 105 treatment trials that reported including both men and women, representing 42,355 participants; 38.6% of the participants were women. Results were fairly similar (34.7% of the participants were women) for the prevention studies. Age distribution and sex were reported in more than 90% of the trials (both treatment and prevention), yet race or ethnicity were reported in only 35.1% of the included treatment trials and 53.6% of the prevention trials. Within the 57 treatment trials (with 45,815 participants) which reported race or ethnicity, 10.5% of participants were reported as African American, and fewer than 1% were reported as Hispanic, Asian, or Native American. There was even less racial and ethnic diversity within the prevention trials: only 5.5% of participants were African American, 1.7% were Hispanic, and fewer than 1% were Asian or Native American.[1]

Knowledge that there were health disparities and efforts to reduce them was not a new topic of discussion. Over a century ago, W. E. B. Du Bois explained how the health status of African Americans was a consequence of social determinants (eg, poverty, inferior education, racism).[2] Following up on this report, Booker T. Washington initiated the Negro Health Improvement Week, a highly successful health education campaign.[3] Until 1985, when the Report of the

Corresponding author: G. Marie Swanson, PhD, Richard M. Fairbanks School of Public Health, Indiana University–Purdue University Indianapolis, 714 N. Senate Avenue, Indianapolis, IN 46202; Fax: (317) 274-3443; E-mail: swanson3@iupui.edu

[1]Richard M. Fairbanks School of Public Health, Indiana University–Purdue University Indianapolis, Indianapolis, Indiana; [2]University of Chicago, Pritzker School of Medicine, Chicago, Illinois.

We thank Stephanie J. Hawley for data abstraction and careful review of each study included in this report.

DOI: 10.1002/cncr.28168, **Received:** March 4, 2013; **Revised:** April 15, 2013; **Accepted:** April 16, 2013, **Published online** May 14, 2013 in Wiley Online Library (wileyonlinelibrary.com)

Secretary's Task Force on Black and Minority Health was released, discussions of health disparities were intermittent. The 1985 Report, which was published in 10 volumes, refocused attention on this topic by providing documentation that African Americans, Hispanics, Native Americans, and Asian/Pacific Islanders experienced excess deaths from many conditions, including cancer, diabetes, unintentional injuries, and infant mortality. The report concluded that there was a serious need for new programs and policies.[4,5]

Ambitious plans, policies, programs, and initiatives followed the re-release of this Report, and although it is not possible to describe all of them, it can be said that health disparities have been a topic of interest for more than a century and that, since 1985, the federal government has placed a strong focus on reducing these disparities. The Centers for Disease Control and Prevention (CDC) and the National Institutes of Health (NIH) have sponsored a wide range of programs directed toward improving diversity in human health research. In 1999, the CDC launched Racial and Ethnic Approaches to Community Health and, in 2000, The Minority Health and Health Disparities Research and Education Act (legal code 42 U.S.C. 202) placed the NIH as the lead federal agency responsible for promoting research to reduce disparities.[6] The NIH Office of Research on Minority Health, created in 1986, was elevated to become The National Center on Minority Health and Health Disparities (NCMHD).[7] The budget for NCMHD increased from the $130 million allocated in 2001 to $195 million in 2006[8] and, over time, funding for health disparities research from all the institutes and centers increased from $2.4 billion in 2003 to approximately $2.6 billion in 2006.[9]

Research funding initiatives helped create, in 2000, 18 Special Populations Networks (SPN) within the National Cancer Institute, leveraging over $20.5 million for health disparities research and cancer awareness and control activities. The goal of these centers, in addition to promoting inclusion of minorities in clinical trials, was reducing the unequal burden of cancer and training the next generation of cancer and cancer disparities researchers. By 2001, these SPNs were housed in the newly created Center to Reduce Cancer Health Disparities (CRCHD) and, by 2007, the CRCHD budget was $28,093,000.[10]

Within the Department of Health and Human Services (DHHS), the Agency for Healthcare Research and Quality's Excellence Centers to Eliminate Ethnic/Racial Disparities was launched in 2001 and The Health Resources and Service Administration's Bureau of Primary Health Care's 100% Access and 0 Disparities Campaign was developed (http://www.ahrq.gov/research/findings/factsheets/minority/exceed/index.html). In 2004, the DHHS Council on Health Disparities was established.[5] One of the primary goals of Healthy People 2010 was the elimination of health disparities within the United States.[11] The recent Affordable Health Care Act (ACA) invests in data collection and research efforts along with promoting the NCMHD to Institute status, in order to better study the differences in health care outcomes among racial and ethnic minorities.[12] Additionally, the ACA encourages the Office on Women's Health to set objectives that related to improvement in women's issues.[13]

The decade since the first report is one in which a large number of federal initiatives have focused on the inclusion of women and minorities in health research. Thus one could reasonably expect that these initiatives would have increased diversity of participants in cancer and other clinical trials, aside from the most recent initiative, the ACA.

Because there has been continued focus at the federal level with funding made available to develop programs to address disparities and include women and racial and ethnic minorities in clinical trials, 2001-2010 was an appropriate point to reassess the state of inclusion of minorities and women in cancer clinical trials, by reviewing the most recent decade of cancer clinical trials. The focus is on cancer clinical trials for 2 reasons: (1) to update the assessment performed for the previous decade by 2 of us (J.C.B., G.M.S.); and (2) to understand some of the underlying reasons for the well-known racial/ethnic and sex disparities in mortality and survival for many cancers.

MATERIALS AND METHODS

The primary objective of this study was to determine the changes in the inclusion of minorities and women in cancer treatment and prevention clinical trials between 1990-2000 and 2001-2010. Criteria for the inclusion of reports in this study are: the article described the primary results for a phase 3 Cancer Treatment or Prevention trial that was reported between 2001 and 2010; contained $\geq$100 participants; had a United States enrollment component; and participants were adults ($\geq$18 years old). Additionally, for this study, articles were excluded if they were not available electronically. To enable comparisons between the 2 reports, the inclusion criteria for the present study are nearly identical to those of the original publication.

OVID was the primary search engine. The search initially yielded 1686 articles. This approach identified a low

10970142, 2013, 16, Downloaded from https://acsjournals.onlinelibrary.wiley.com/doi/10.1002/cncr.28168 by Cochrane Canada Provision, Wiley Online Library on [03/08/2023]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

number of prevention studies. We therefore undertook an additional search for eligible prevention trials, which yielded another 375 articles. One of the study authors screened, each of the 2051 articles for inclusion in the study and a second author verified the acceptability for inclusion. Any differences were resolved by discussion and agreement.

An ACCESS database and data entry form was developed to record and store variables for the analysis. The following data were extracted from each article: general publication characteristics, including authors, funding source, and journal name; study characteristics, including sample size, years of enrollment, and whether it was a single-site or multisite trial; and information about selection criteria, including age, sex, race, cancer site, cancer cell type, specific behaviors, and hormone status. Additionally, information about outcome measures such as survival status, recurrence, quality of life, toxicity, treatment response, disease progression, knowledge status, and specific behaviors was extracted. Information about predefined subgroup analyses; and whether or not the authors reported characteristics of study participants, including age, sex, race, treatment characteristics, and selected behavioral variables also was captured.

All data were entered directly into the ACCESS database via a data form, and 100% quality control was conducted to ensure accuracy of the data entered. The majority of data were extracted and entered by 1 individual, and G.M.S. conducted the quality control. For a minority of the articles, G.M.S. reviewed and entered articles into the database, and K.K. conducted the quality control assessment. Any differences were resolved by discussion and agreement. Microsoft Excel was used to organize and summarize the information. Descriptive statistics are used below to present the findings.

RESULTS

A total of 277 cancer treatment trials and 27 cancer prevention trials published from 2001 through 2010 that met the inclusion criteria were identified and were included in the analysis dataset. A total of 191,199 patients were reported in the treatment trials and 138,472 patients were reported in the prevention studies. The number of participants in each trial was distributed differently between the decades. In 2001-2010, 18.1% of the treatment trials had greater than 1000 participants compared with only 10.2% for trials published between 1990 and 2000. Forty-six different journals published treatment trials in 2001-2010, with the majority (54.9%) published in the *Journal of Clinical Oncology,* followed by *Cancer* (5.4%) and *The New England Journal of Medicine* (5.1%). In contrast, only 16 journals published prevention studies, with the *American Journal of Public Health* (14.8%) and *Preventive Medicine* (14.8%) publishing the most articles. A similar range of journals published these types of articles in the previous decade.

During the recent decade, 50.4% of the treatment trials were funded by pharmaceutical companies and 35.9% were funded by the NIH. This is different from the period 1990-2000, in which the majority of treatment trials were funded by the NIH. Of all reported prevention studies in the recent decade, 78% were funded by the NIH compared with 67.9% in 1990-2000. Pharmaceutical companies did not fund any prevention studies in either decade.

Detailed trial characteristics for this and the previous decade are shown in Tables 1 and 2.

Reporting of Race/Ethnicity

Of treatment studies published from 2001 to 2010, 51.6% of studies, with 104,337 participants, reported on race/ethnicity, compared with 35.1% of treatment studies from 1990 to 2000 with 45,815 participants. Additionally, in 2001-2010, when race/ethnicity was reported, 82.9% were white, 6.2% were African American, 3.3% were Asian, 2.2% were Hispanic, 0.1% were Native American, and 5.4% were characterized as Other. This is in contrast to 1990-2000, during which 89% were white, 10.5% were African American, 0.4% were Hispanic, and 0.04% were Asian (Table 3).

Publications of prevention studies reported on race/ethnicity at a greater frequency than treatment studies, at 77.8% from 2001-2010. From 1990-2000, prevention studies reported this information 53.6 percent of the time. Prevention trial participants from 2001-2010 continued to include whites as the predominant race/ethnicity group, even though the percentage dropped from 92.5% in 1990-2000, compared to the 81.5% in 2001-2010. More African Americans participated from 2001-2010 (11.6%) than in the previous decade (5.5%). Additional enrollment characteristics of the participants in studies which included race/ethnicity information are presented in Table 3.

Reporting of Sex

Numbers of women were reported at similar frequencies from 2001 to 2010 in both treatment studies (95%) and prevention trials (93%). In 1990-2000, the number of women included in treatment studies was reported 92.7% of the time. Women accounted for 40.2% of the participants in treatment trials, which were described as

10970142, 2013, 16, Downloaded from https://acsjournals.onlinelibrary.wiley.com/doi/10.1002/cncr.28168 by Cochrane Canada Provision, Wiley Online Library on [03/08/2023]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 1. Characteristics of Cancer Clinical Treatment Trials Published

Characteristic	2001-2010	1990-2000
Total study participants	191,199	
No. of study participants included in publication		
100-499	146 (52.7)	155 (75.6)
500-999	81 (29.2)	29 (14.2)
≥1000	50 (18.1)	21 (10.2)
Year published		
2001	15 (5.4)	
2002	17 (6.1)	
2003	21 (7.6)	
2004	21 (7.6)	
2005	25 (9.0)	
2006	33 (11.9)	
2007	23 (8.3)	
2008	45 (16.2)	
2009	34 (12.3)	
2010	43 (15.5)	
Total studies published	277	
Year participants first entered the trial		
1974-1984	2 (0.9)	
1985-1990	14 (6.0)	
1991-1995	42 (18.0)	
1996-2000	81 (34.8)	
2001-2005	84 (36.1)	
2006-2010	10 (4.3)	
Journal in which treatment trial was published[a]		
Journal of Clinical Oncology	152 (54.9)	138 (53.3)
The New England Journal of Medicine	14 (5.1)	31 (11.9)
Cancer	15 (5.4)	20 (7.7)
The Lancet	13 (4.7)	NR
Annals of Oncology	14 (5.1)	NR
Journal of Urology	8 (2.9)	NR
American Journal of Clinical Oncology	3 (1.1)	NR
Gynecologic Oncology	7 (2.5)	NR
International Journal of Radiation Oncology, Biology, Physics	13 (4.7)	N NR
Journal of the National Cancer Institute	5 (1.8)	NR
Trial funding source		
NIH or other federal funding agency only	94 (35.9)	126 (61.5)
Pharmaceutical company only	132 (50.4)	27 (13.2)
Other only	3 (1.1)	NR
NIH or other federal funding agency and other	7 (2.7)	14 (6.8)
NIH or other federal funding agency and pharmaceutical company	20 (7.6)	13 (6.3)
NIH or other federal funding agency and pharmaceutical company and other	4 (1.5)	NR
Other and pharmaceutical company	2 (0.8)	NR
Not reported	19 (7.3)	16 (7.8)

Data are presented as no. (%).
Abbreviations: NIH, National Institutes of Health; NR, not reported.
[a] Listed if ≥3 articles.

enrolling both sexes from 2001 to 2010, compared with encompassing 38.6% of the participants in 1990-2000.

In prevention studies reported during 2001-2010, women were included 26.5% of the time versus 34.7% from 1990-2000. Women more frequently participated in treatment studies, as 45.5% of treatment articles reported that >50% of the participants were women, compared with 15.6% in treatment studies in the most recent decade. In 1990-2000, 11.5% of prevention studies and 9.5% of treatment studies reported that more than 50% of participants were women. Additional enrollment characteristics of the studies which included sex information are presented in Table 3.

TABLE 2. Characteristics of Cancer Clinical Prevention Trials Published

Characteristic	2001-2010	1990-2000
Total study participants	138,472	
No. of study participants included in publication		
100-499	8 (30)	NR
500-999	7 (26)	NR
≥1000	12 (44)	NR
Year published		
2001	0 (0.0)	
2002	1 (3.7)	
2003	2 (7.4)	
2004	2 (7.4)	
2005	6 (22.2)	
2006	2 (7.4)	
2007	2 (7.4)	
2008	4 (14.8)	
2009	4 (14.8)	
2010	4 (14.8)	
Total studies published	27	
Year participants first entered the trial		
1974-1984	0 (0.0)	
1985-1990	1 (6.3)	
1991-1995	4 (25.0)	
1996-2000	4 (25.0)	
2001-2005	6 (37.5)	
2006-2010	1 (6.3)	
Journal in which prevention trial was published[a]		
Preventive Medicine	4 (14.8)	NR
American Journal of Public Health	4 (14.8)	NR
JAMA	3 (11.1)	NR
Trial funding source		
NIH or other federal funding agency	21 (78)	38 (67.9)
Pharmaceutical company	0 (0)	NR
American Cancer Society	4 (15)	(8.9)
University or medical center	1 (4)	NR
Other	1 (4)	NR

Data are presented as no. (%).
Abbreviations: NIH, National Institutes of Health; NR, not reported.
[a] Listed if ≥3 articles.

Reporting of Selection Criteria

Most commonly, cancer site, disease stage, performance status, cell type, age, and sex were identified as selection criteria for treatment studies from 2001 to 2010. Additional variables were collected and reported in the current study, thus comparative data are not available for all variables. Performance score and sex were most frequently used as selection criteria in 1990-2000. Additionally in this

10970142, 2013, 16, Downloaded from https://acsjournals.onlinelibrary.wiley.com/doi/10.1002/cncr.28168 by Cochrane Canada Provision, Wiley Online Library on [03/08/2023]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 3. Enrollment Characteristics of Studies which Included Minorities and Women

Enrollment Characteristic	2001-2010	1990-2000
Treatment trials		
Articles that reported race/ethnicity	143 (51.6)	57 (35.1)
Number of participants included when race/ethnicity information was reported	104,337	45,815
White	86,484 (82.9)	40,803 (89.0)
African American	6403 (6.1)	4811 (10.5)
Hispanic	2333 (2.2)	183 (0.4)
Asian	3398 (3.3)	18 (0.04)
American Indian	79 (0.1)	NR
Other	5640 (5.4)	NR
Articles with no African American participants	31 (21.7)	NR
Articles with no Hispanic participants	76 (53.1)	51 (89.5)
Articles with no Asian participants	82 (57.3)	56 (98.2)
Articles with no American Indian participants	126 (88.1)	56 (98.2)
Articles that reported sex	263 (95.0)	190 (92.7)
Women	42,957 (40.2)	16,349 (38.6)
Men	63,950 (59.8)	26,006 (61.4)
Articles with >50% of participants who were women	26 (15.6)	10 (9.5)
Prevention trials		
Articles that reported race/ethnicity	21 (77.8)	17 (53.6)
Number of participants included when race/ethnicity information was reported	91,663	91,741
White	74,695 (81.5)	84,860 (92.5)
African American	10,624 (11.6)	5046 (5.5)
Hispanic	3294 (3.6)	1560 (1.7)
Asian	65 (0.1)	275 (0.3)
American Indian	1 (0.0)	14 (0.01)
Other	2984 (3.3)	NR
Articles with no African American participants	6 (28.6)	NR
Articles with no Hispanic participants	13 (61.9)	10 (58.8)
Articles with no Asian participants	20 (95.2)	15 (88.7)
Articles with no American Indian participants	20 (95.2)	NR
Articles that reported sex	25 (93.0)	52 (92.9)
Women	7790 (26.5)	25,676 (34.7)
Men	9874 (33.6)	48,319 (65.3)
Articles with >50% of participants who were women	5 (45.5)	3 (11.5)

Data are presented as no. (%).
Abbreviation: NR, not reported.

TABLE 4. Cancer Clinical Treatment Trials: Participant Selection Criteria

Selection Criteria	2001-2010	1990-2000
Age	139 (50.2)	60 (29.3)
Sex	110 (39.7)	91 (44.4)
Men	26 (9.4)	NR
Women	84 (30.3)	NR
Cancer site	258 (93.1)	NR
Stage of disease at diagnosis	242 (87.4)	NR
Race	1 (0.4)	NR
Various treatment characteristics	267 (96.4)	NR
Performance status or score	213 (76.9)	122 (59.5)
Cell type of cancer	180 (65.0)	NR
Menopause status	20 (7.2)	14 (6.8)
Weight gain or loss	5 (1.8)	9 (4.4)
Weight	2 (0.7)	NR
Pregnancy status	77 (27.8)	9 (4.4)
Smoking habits	3 (1.1)	NR
Education level	0 (0.0)	NR
Occupation	0 (0.0)	NR
Marital status	0 (0.0)	NR
Dietary habits	2 (0.7)	NR
Hormonal status	42 (15.2)	NR

Data are presented as no. (%).
Abbreviation: NR, not reported.

TABLE 5. Cancer Clinical Prevention Trials: Participant Selection Criteria

Selection Criteria	2001-2010	1990-2000
Age	19 (70.4)	41 (73.2)
Sex	16 (59.3)	21 (37.5)
Men	3 (11.1)	NR
Women	13 (48.1)	NR
Cancer site	6 (22.2)	NR
Stage of disease diagnosis	0 (0.0)	NR
Race	5 (18.5)	5 (8.9)
Various treatment characteristics	7 (25.9)	NR
Performance status or score	1 (3.7)	NR
Cell type of cancer	3 (11.1)	NR
Menopause status	1 (3.7)	3 (5.5)
Weight gain or loss	0 (0.0)	NR
Weight	0 (0.0)	6 (10.7)
Pregnancy status	3 (11.1)	7 (12.5)
Smoking habits	1 (3.7)	14 (25.0)
Education level	2 (7.4)	3 (5.4)
Occupation	0 (0.0)	8 (14.3)
Marital status	0 (0.0)	NR
Dietary habits	1 (3.7)	8 (16.1)
Hormonal status	1 (3.7)	NR

Data are presented as no. (%).
Abbreviation: NR, not reported.

decade, sex was more often (44.4%) used to select participants compared with 2001-2010 (39.7%). Except for 1 study, sex was an explicit selection criterion only in studies of sex-specific malignancies, with 74 studies of carcinoma of the female breast, cervix, ovary, or endometrium and 17 studies of prostate or testicular carcinoma. Race and ethnicity were not included in Table 2 because no report stated that they were criteria for participant selection. When looking at prevention studies, age and sex were most often identified as selection criteria in both decades. Tables 4 and 5 provide additional details on participant selection criteria.

Reporting of Outcome Measures

The most frequently reported outcome studied in treatment trials in both decades was survival (79.8% [2001-2010], 87.3% [1990-2000]). Other variables that were frequently reported in both decades included treatment

10970142, 2013, 16, Downloaded from https://acsjournals.onlinelibrary.wiley.com/doi/10.1002/cncr.28168 by Cochrane Canada Provision, Wiley Online Library on [03/08/2023]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

response (63.2% [2001-2010], 72.2% [1990-2000]) and disease progression/regression (61.4% [2001-2010], 36.6% [1990-2000]). Toxicity was reported much more frequently in 1990-2000 as an outcome variable (87.3% in 1990-2000 versus 22% in 2001-2010. This finding is likely due to data classification differences between the original 2002 study conducted by Swanson and Bailar and the current report. The current study classified toxicity as a primary outcome variable only if the study reported having adequate power to detect toxicity-related outcomes. Recurrence (10.8% [2001-2010], 20.5% [1990-2000]) and quality of life (27.1% [2001-2010]; 15.1% [1990-2000]) were reported less often in both decades.

Sixty-three percent of the cancer prevention studies in the recent decade reported a specific prevention behavior as the most frequent outcome under study (eg, dietary changes, use of sunscreen) to lessen risk of a particular cancer. The original study collapsed the outcome variables for prevention studies into categories, rather than delineating specific behaviors. Consequently, accurate direct comparisons that can be made are limited.

Reporting of Subgroup Analyses

During the period 2001-2010, subgroup analyses were presented in 17.3% of treatment trials and 14.8% of prevention studies. Women were specified as a subgroup 7.6% of the time in treatment articles, compared with 6.3% in 1990-2000. In 3.7% of the cancer prevention studies published from 2001 to 2010, women were reported in subgroup analyses, compared with 19.6% in 1990-2000. When subgroup analyses were presented in the primary publication, race/ethnicity was specified in 4.7% of cancer treatment trial reports from 2001 to 2010, compared with 2.4% in 1990-2000. Cancer prevention studies specified race/ethnicity as a particular subgroup in 11.1% of the reports from 2001-2010 compared with 8.9% in 1990-2000.

DISCUSSION

The purpose of this investigation was to determine whether the reported inclusion of minorities and women in cancer clinical trials changed between 1990-2000 and 2001-2010. Although there was a 16.5% increase in the reporting of race/ethnicity in cancer treatment trials in 2001-2010 compared with 1990-2000, this information is still poorly reported, with only 51.6% of all trials presenting this information in major scientific journals. The picture is better in cancer prevention studies, with 77.8% of publications reporting on race/ethnicity in 2001-2010 compared with 53.6% in 1990-2000. NIH guidelines, amended in 2001, for reporting on inclusion of women and minorities in clinical research have yet to achieve the intended goals.[14] Although the majority of the cancer treatment trials were funded by pharmaceutical companies, the Food and Drug Administration has adopted similar rules for reporting demographic information. These results are therefore perplexing.

Diversity of race/ethnic groups in cancer clinical trials has improved very slightly. The percentage of whites included in cancer treatment and prevention trials studies dropped the last decade to 6% and 11%, respectively. However, whites still make up the significant majority (>80% of all participants in cancer clinical trials). The number of African Americans participating in cancer treatment trials actually declined over the previous 10 years, while there were slight increases in the inclusion of other minority groups. Within cancer prevention studies, African Americans represented 11.6% of all participants, with other minorities each representing $\leq$4%. Other reviews have reported similar results on the reporting and inclusion of minorities in clinical trials.[15–18]

Sex was reported at high rates in both decades; however, the actual inclusion of women remains low, making up only 40.2% of participants in cancer treatment trials and 26.5% in cancer prevention trials (excluding sex-specific trials). This finding is supported by recent meta-analyses that discuss the low inclusion rates of women in clinical research.[19–21] It is well known that cancer treatment is not a one-size-fits-all approach. However, the scientific practice underlying the study of the effectiveness of cancer treatment and prevention has not changed. An improved understanding of the differences between subgroups is critical to improving the risk/benefit profile for a wide-range of chemotherapeutics and prevention efforts. Without this information, the ability to generalize results to different subgroups and to effectively manage patients is substantially reduced. This limits the ultimate utility of both science and practice based on findings of cancer clinical trials carried out over this decade. With minimal reporting of this information, the ability to develop hypotheses for future cancer treatment or prevention trials is limited for sex and racial/ethnic subgroups.[19]

This report shows that improvements proposed a decade ago in the design and conduct of clinical trials have yet to be achieved or, at a minimum, they are not reported in the primary publications. Swanson and Bailar offered specific and detailed solutions to the scientific and social equity dilemmas of cancer clinical trials

10970142, 2013, 16, Downloaded from https://acsjournals.onlinelibrary.wiley.com/doi/10.1002/cncr.28168 by Cochrane Canada Provision, Wiley Online Library on [03/08/2023]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

practice a decade ago.[1] We reiterate some of those recommendations here. First, journal editors should require that basic demographic information be reported in any primary trial report. Second, the same scientific precision that is applied to randomization should be applied to participant selection and recruitment. For example, statistical precision of results for subgroups depends on the numbers in those groups, thus oversampling of a particular ethnic group may be necessary to ensure results are generalizable for the intended treatment groups. Third, the rationale for specific inclusion or exclusion of women and minorities should be described in the report so that readers can understand required trade-offs and enrollment, as it may not always be appropriate to include all subgroups. Finally, it is insufficient to require only reporting of the information; attention also should be paid to increasing the access of trials to women and minorities. NIH funding to train health researchers who are women and who are from racial/ethnic minorities is an essential approach to improving the quality of science in clinical trials.

Clinical trials are conducted over many years, and this report evaluates published studies. Therefore, it may be that studies newer than the ones available in the current scientific literature have a different experience than we can observe at this time. These are important components of study design, so it is critical that recommendations such as ours are followed rigorously. Because recent studies are funded predominantly by pharmaceutical companies, it also may be appropriate for federally funded trials or those trials submitted to the US Food and Drug Administration (FDA) for clearance to apply a certain level of due diligence to recruitment practices and to require that information to be reported and assessed on a specified basis. It may further be appropriate for the FDA to grant approval for the use of pharmaceuticals only for the sexes and races/ethnicities included in the clinical trials upon which the population use of these drugs is recommended.

There are limitations to this systematic review. Although the intent was to include all adult primary reports of phase 3 cancer clinical treatment and prevention studies that were published from 2001 to 2010 and included United States enrollment, some manuscripts may have been missed due to the use of the selected search engines. Nonetheless, it seems unlikely that the results would have been altered significantly, because others have found similar distributions in other therapeutic and prevention areas or funding mechanisms. Furthermore, because some of the patients were enrolled in the early part of the decade, it may have been too early to see the impact of some policy changes.

In conclusion, in order to improve treatment for cancer patients and prevention strategies for those at risk, a critical reassessment of the strategies used to improve inclusion and reporting of minorities and women must be performed, because many valuable published studies have limited application to diverse populations. It also important to be clear that the science of clinical trials remains far less rigorous than what is needed and what can be achieved, because limited attention is given to defining, recruiting, and reporting on selection of study participants.

FUNDING SOURCES

This study was funded by Indiana University–Purdue University Indianapolis and the Indiana University Simon Cancer Center.

CONFLICT OF INTEREST DISCLOSURES

The authors made no disclosures.

REFERENCES

1. Swanson GM, Bailar JC, 3rd. Selection and description of cancer clinical trials participants—science or happenstance? *Cancer.* 2002; 95:950-959.
2. Du Bois, WEB. The health and physique of the Negro American. *Am J Public Health.* 1906;93:272-276
3. Quinn SC, Thomas, SB. The National Negro Health Week, 1915-1951: a descriptive account. *Wellness Perspectives.* 1996;12:172-179.
4. Orsi JM, Margellos-Anast H, Whitman S. Black-White health disparities in the United States and Chicago: a 15-year progress analysis. *Am J Public Health.* 2010;100:349-356.
5. Thomas SB, Benjamin GC, Almarios D, Lathan, M. Historical and current policy efforts to elminate racial and ethnic health disparities in the United States: future opportunities for public health education. *Health Promot Prat.* 2006;7:324-330.
6. Beal A. Policies to reduce racial and ethnic disparities in child health and health care. *Health Aff (Millwood).* 2004;23:171-179
7. Thomson G, Mitchell F, Williams M. Examining the Health Disparities Research Plan of the National Institutes of Health: Unfinished Business. Washington, DC: The National Academies Press; 2001.
8. National Institutes of Health. Annual performance plan and report. Available at: http://dpcpsi.nih.gov/opep/documents/Perform/FY_2006_NIH_Performance_Plan.pdf. Accessed November 8, 2012.
9. National Institute of Minority Health and Health Disparities. NIH health disparities strategic plan. Available at: http://www.nimhd.nih.gov/about_ncmhd/index2.asp. Accessed November 8, 2012.
10. Springfield S. Catalyzing transdisciplinary regional partnerships to eliminate cancer health disparities. Sharing in our vision: center to reduce cancer health disparities. Available at: http://www.cancermeetings.org/chdsummit07/PDF/Springfield.pdf. Accessed August 20, 2012.
11. US Department of Health and Human Services. Healthy People 2010. Available at: http://www.healthypeople.gov/2010/. Accessed September 16, 2012.
12. Henderson A, Robinson W, Finegold K. The Affordable Care Act and Latinos. Washington, DC: Department of Health and Human Services; 2012.
13. Cuellar A, Simmons A, Finegold K. The Affordable Care Act and Women. Washington, DC: Department of Health and Human Services; 2012.

10970142, 2013, 16, Downloaded from https://acsjournals.onlinelibrary.wiley.com/doi/10.1002/cncr.28168 by Cochrane Canada Provision, Wiley Online Library on [03/08/2023]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

14. National Institutes of Health. NIH policy and guidelines on the inclusion of women and minorities as subjects in clinical research. Available at: http://grants.nih.gov/grants/funding/women_min/guidelines_amended_10_2001.htm. Accessed June 17, 2012.
15. Berger JS, Melloni C, Wang TY, et al. Reporting and representation of race/ethnicity in published randomized trials. *Am Heart J.* 2009; 158:742-747.
16. Park IU, Taylor AL. Race and ethnicity in trials of antihypertensive therapy to prevent cardiovascular outcomes: a systematic review. *Ann Fam Med.* 2007;5:444-452.
17. Hutchins LF, Unger JM, Crowley JJ, Coltman CA Jr, Albain KS. Underrepresentation of patients 65 years of age or older in cancer-treatment trials. *N Engl J Med.* 1999;341:2061-2067.
18. Murthy VH, Krumholz HM, Gross CP. Participation in cancer clinical trials: race-, sex-, and age-based disparities. *JAMA.* 2004;291: 2720-2726.
19. Geller SE, Koch A, Pellettieri B, Carnes M. Inclusion, analysis, and reporting of sex and race/ethnicity in clinical trials: have we made progress? *J Womens Health (Larchmt).* 2011;20:315-320.
20. Kim E, Carrigan TP, Menon V. Enrollment of women in National Heart, Lung, and Blood Institute-funded cardiovascular randomized controlled trials fails to meet current federal mandates for inclusion. *J Am Coll Cardiol.* 2008;52:672-673.
21. Blauwet LA, Hayes SN, McManus D, Redberg RF, Walsh MN. Low rate of sex-specific result resporting in cardiovascular trials. *Mayo Clin Proc.* 2007;82:166-170.

10970142, 2013, 16, Downloaded from https://acsjournals.onlinelibrary.wiley.com/doi/10.1002/cncr.28168 by Cochrane Canada Provision, Wiley Online Library on [03/08/2023]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License